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Spiral galaxies of the Virgo cluster: possible black hole seeding/feeding of the bulgeless spiral galaxy NGC 4424 via an infalling star cluster

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ABSTRACT

A natural process by which galaxies grow is through their mutual gravitational attraction and subsequent unification. Orbiting a regular galaxy, the main body of a petite galaxy can be stripped away, and it will be eroded further as it intertwines deeper during the merger process. However, the heart of the infalling galaxy, represented by its more tightly-bound central star cluster, is more resilient. We have discovered a red, tidally-stretched, star cluster positioned 4″.9 (∼390 pc in projection) from, and pointing towards, the centre of the post-merger spiral galaxy NGC 4424. The star cluster, which we refer to as 'Nikhuli', has a near-infrared luminosity of $6.9^{+2.0}_{-1.5} \times 10^7\ L_{\odot,F160W}$, and likely represents the core of a captured/wedded low-mass early-type galaxy. Moreover, Nikhuli contains a high-energy X-ray point-source, likely signalling the presence of an active black hole, with a spectrum consistent with a power-law slope $\Gamma = 1.7$ and an X-ray luminosity $L_{0.5-8\,\mathrm{keV}} = 6.31^{+7.50}_{-3.77} \times 10^{38}$ erg s^{-1} (90% confidence). Lacking an outward-pointing comet-like appearance, the stellar structure of Nikhuli favours infall rather than an ejection from a gravitational-wave recoil event. The stellar mass and velocity dispersion of NGC 4424 predict a central black hole of (0.6–1.0)×10^5 M$_\odot$, similar to the predicted intermediate-mass black hole in Nikhuli, and thus suggestive of a black feeding mechanism for bulgeless late-type galaxies. We may potentially be witnessing black hole seeding by capture and sinking, with external star clusters the delivery vehicle.

Key words: X-rays: individual: NGC 4424 X-3 – galaxies: spiral – galaxies: star clusters: general – X-rays: galaxies

1 INTRODUCTION

Galaxies have long been known to merge, with smaller galaxies showering and stoking bigger galaxies (Lindblad 1927; Spitzer & Baade 1951; Baade & Minkowski 1954; Vorontsov-Velyaminov 1959; Ibata et al. 2019b). The dense and compact nuclear star clusters — which are common features at the centres of low- and intermediate-mass galaxies (e.g. Sandage et al. 1985; Böker et al. 2002; Graham & Guzmán 2003; Balcells et al. 2003, 2007) — are more strongly gravitationally-bound than their host galaxies, and are thus more able to resist disruption during the cannibalistic process (e.g., Bekki et al. 2001, 2003). Indeed, the dual and multiple nuclei of massive early-type galaxies is a testament to the robustness of these captured star clusters during galaxy assimilation (e.g., Jenner 1974; Petrosian et al. 1978; Koroviakovskii et al. 1981; Tonry 1985; Blakeslee & Tonry 1992; Bonfini & Graham 2016). Such off-centre nuclei are, however, harder to spot in late-type galaxies, due in part to confusion with knots of star-formation, with the later stages of the merger/digestion process increasingly difficult to detect. Although, this is changing with the discovery of

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faint, drawn-out debris streams of former galaxies (Putman et al. 1998; Belokurov et al. 2006; Martin et al. 2014; Drlica-Wagner et al. 2015; Shipp et al. 2018). As is the case with the early-type galaxies, the detection of a remnant nuclear star cluster near the centre of a late-type galaxy reveals that the Roche-lobe destruction and tidal-shredding of one's galactic neighbours does not just build a galaxy's diffuse halo, but also the galaxy proper.

From deep *Chandra X-ray Observatory* (*CXO*) exposures of 100 early-type galaxies in the Virgo cluster, half of which possess a nuclear star cluster (Côté et al. 2006; Ferrarese et al. 2006), Gallo et al. (2010) have effectively shown that nuclear star clusters tend not to contain X-ray point-sources. From a subsample of 30 low-mass, mostly-nucleated, galaxies expected to harbour an intermediate-mass black hole (IMBH), Graham & Soria (2019) reported that just 3 of these contained a central X-ray point-source, therefore ruling out stellar-mass X-ray binaries as a common feature of nuclear star clusters. In contrast, studying the Virgo cluster's late-type galaxies with star formation rates above $\sim$0.3 $M_\odot$ yr^{-1}, and thus with fuel to ignite active galactic nuclei (AGN), Graham et al. (2021) have discovered a hotbed of activity. As a part of the *Chandra X-ray Observatory* (*CXO*) Large Project, titled 'Spiral galaxies of the Virgo Cluster' (PI: R.Soria, Proposal ID: 18620568), Graham et al. (2019) have reported on 33 IMBH candidates in low-mass late-type spiral galaxies, and Graham et al. (2021) have found a central X-ray point source in 14 of these. The presence of an X-ray point-source in a nuclear star cluster is, therefore, more likely to signal the presence of a massive black hole rather than an accreting stellar-mass object. For all but one of the half dozen systems for which an X-ray spectrum could be obtained, it was found to be consistent with a power-law rather than a blackbody curve expected from the accretion disk around a compact, stellar-mass object.

Among the Virgo Cluster spirals in our sample, the SB(s)a galaxy NGC 4424 is of particular interest. While there is no detectable X-ray point-source at the centre of NGC 4424, there is a nearby (5″) 'off-centre' source, NGC 4424 X-3. First publically reported by Boselli et al. (2018), they suggested it is either an X-ray binary or the nucleus of a merged galaxy. Two brighter X-ray point-sources plus a fainter fourth point-source reside in the outskirts of this galaxy. Although Feng & Soria (2011), Bachetti et al. (2014) and Kaaret et al. (2017) reveal that super-Eddington accretion (Alexander & Natarajan 2014; Soria et al. 2014) is thought to explain many of the off-centre ultraluminous X-ray sources ($L_X \approx 3 \times 10^{39}$ to 10^{41} ergs s^{-1}), some might be IMBHs from accreted galaxies, in particular the hyperluminous X-ray sources (HLXs: $L_X > 10^{41}$ erg s^{-1}, Gao et al. 2003) such as ESO 243-49 (Farrell et al. 2009; Soria et al. 2010) and those at the centres of dense star clusters. Here, we report the association of NGC 4424 X-3 ($L_{0.5-8\,\mathrm{keV}} \approx 6 \times 10^{38}$ erg s^{-1}) with a red, elongated star cluster. In Section 2, we provide some background information and references to NGC 4424 before presenting optical and near-infrared images along with an X-ray map. We perform our own re-analysis of the *Hubble Space Telescope* (*HST*) image and the *CXO* data, and provide an extended discussion. Section 3 offers a prediction for both the mass of the central black hole in NGC 4424 and for the suspected incomer. A discussion, centred around off-centred sources, is provided in Section 4.

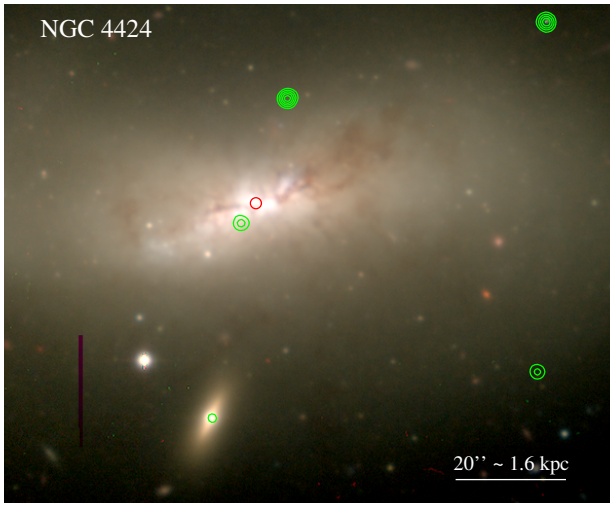

Figure 1. Next Generation Virgo Cluster Survey (NGVS: Ferrarese et al. 2012) image of the post-merger galaxy NGC 4424 (red = i filter; green = g; blue = u), with *CXO*/ACIS-S contours (0.5–7.0 keV band) overlaid in green. North is up, east is to the left. The red circle shows the NED-provided position for the galaxy's optical nucleus, with a radius of 1 arcsec capturing the associated uncertainty. The near-central, but off-centre, X-ray point-source (NGC 4424 X-3) is associated with an elongated star cluster (see Figure 2). The galaxy to the south is IC 3366, a background galaxy at a distance of $\sim$110 Mpc, whil ethe vertical red stripe is an artifact that should be ignored.

2 THE OBSERVATIONS

2.1 Preface

From past observations, NGC 4424 (Figure 1) has been identified as a post-merger galaxy undergoing substantial central star formation (Kenney et al. 1996). Cortés et al. (2006) and Boselli et al. (2018) provide a detailed analysis of NGC 4424, finding that it experienced a gas rich, unequal-mass merger event less than 0.5 Gyr ago. The encounter has fueled an outburst of ionised gas, evident as a $\sim$10 kpc plume pointing in the opposite direction to the $\sim$110 kpc long, (ram pressure)-stripped, tail of HI gas as NGC 4424 ploughs through the hot intracluster medium of the Virgo cluster.

Riess et al. (2016) report a Cepheid-based distance modulus of 31.080±0.292 (D= 16.4 ± 0.8 Mpc) for NGC 4424, which we adpot here due to its liberal error bar. This agrees with Hatt et al. (2018) who report a 'tip of the red giant branch' distance modulus of $31.00 \pm 0.03_{\mathrm{stat}} \pm 0.06_{\mathrm{sys}}$ ($15.8 \pm 0.2_{\mathrm{stat}} \pm 0.4_{\mathrm{sys}}$ Mpc), in good agreement with Munari et al. (2013) who derived a SN Ia distance modulus of 30.95, and Cortés et al. (2008) who reported a value of 30.91 based on the Tully & Fisher (1977) relation.

2.2 Optical/near-IR images

NGC 4424 is a bulgeless galaxy containing a knotty bar-like feature extending to a radius of 14.″4 (with a brighter inner region extending to 7″), plus an exponential-like disc. Using a Wide Field Camera 3 Infrared (WFC3/IR) *F160W H*-band image from *Hubble Space Telescope* (*HST*) Proposal ID 12880 (PI: A.Riess), we have modelled NGC 4424 using the Isofit/Cmodel task (Ciambur 2015), run within the IRAF package ELLIPSE. The image has a PSF Full Width at Half Maximum (FWHM) spatial resolution of 0.″.18, and the near-IR helps to reduce the impact of dust, young stars and avoid H_α emission. The resulting galaxy model has been subtracted in order to highlight the residual structure, which has revealed an elongated, off-centre and off-bar, star cluster ∼5 arcseconds (390 pc) to the south-east of the galaxy centre. This somewhat stretched star cluster is drawn-out in the direction of the centre of the galaxy, and likely represents the nucleus of the shredded galaxy which is in the final digestion stages of merging and assimilation with NGC 4424.

Working from the residual image (galaxy minus model), the elongated source has a flux of 844 ± 29 electrons per second. Using a WFC3/IR F160W zero-point[1] of 25.946 mag (AB photometric system: Koekemoer et al. 2013), this corresponds to an apparent magnitude of 18.63±0.04 mag (AB). Using a distance modulus of 31.08±0.29, gives an absolute magnitude of −12.45±0.29 mag. With an F160W absolute magnitude for the Sun of +4.60 mag (AB mag: Willmer 2018), the stellar luminosity is $6.6^{+2.0}_{-1.6} \times 10^6\, L_{\odot,F160W}$; and assuming an F160W stellar mass-to-light ratio of 0.5 (Graham & Spitler 2009, see their Figure A1), the stellar mass is therefore $3.3 \times 10^6\, M_\odot$.

The point-source NGC 4424 X-3 (Boselli et al. 2018) is spatially coincident with the elongated star cluster (see Figure **??**) which we suggest is the nucleus of the galaxy which merged with NGC 4424, somewhat similar to the simulation presented by (Bellovary et al. 2010) and (Bellovary et al. 2021). The cluster has a mean F814W−F160W colour equal to 1.60±0.10 mag, which is redder than the surrounding galaxy colour of 1.44±0.04 mag, and indicative of an older stellar population. As such, the cluster is not a knot of recent star formation triggered by the past merger event. Given that the cluster also shows up in the F814W image, it rules out the possibility that the cluster might instead be [Fe II] 1.64 μm line emission from an ionized bubble around NGC 4424 X-3.

The centres of galaxies of course represent a unique position, associated with the bottom of the gravitational potential well, where massive objects/structures, such as dense star clusters, tend to sink due to the braking force, and thus orbital decay, arising from 'dynamical friction' (Chandrasekhar & von Neumann 1943; Baranov & Batrakov 1974; Tremaine et al. 1975; Inoue 2011; Arca-Sedda & Capuzzo-Dolcetta 2014). Although, in low-mass galaxies with low Sérsic indices and thus a shallow gradient to the central gravitational potential (e.g., Terzić & Graham 2005; Terzić & Sprague 2007), star clusters are knwon to wander about (Binggeli et al. 2000). The significant stretching of the red star cluster in the direction of the galaxy centre seems to favour an infall scenario with tidal (Roche lobe) stretching of either the cluster or the paltry remains of the accreted galaxy still surrounding its nuclear star cluster. Lacking an outward-pointing head-tail comet-like appearance, we suspect that this hot[2] star cluster is probably not an ejected nucleus — via a gravitational wave recoil event after the merger of two black holes (Blecha & Loeb 2008; Holley-Bockelmann et al. 2008; Komossa & Merritt 2008; Askar et al. 2020). It is also clearly not shaped like a (background) galaxy. Instead, it is likely the seed of a galaxy harvested from the nearby environment and consumed by NGC 4424. We refer to this star system as Nikhuli[3], a name which relates to a festive period of celebrating and wishing for a rich harvest.

2.3 X-ray data

2.4 Chandra X-ray Observatory data

Using the 14.87 ksec *CXO* Advanced CCD Imaging Spectrometer (ACIS) exposure (ObsID: 19408. PI: Soria) taken on 2017-04-17, Boselli et al. (2018, see their section 4.2) reported on the obscured X-ray source NGC 4424 X-3 located 4.9 arcseconds (∼390 pc) to the south-east of the nucleus. As noted, NGC 4424 X-3 is coincident with the star cluster. From the *CXO* image, we observed 8 X-ray counts, of which none were in the soft band (0.5-1 keV), 1 was in the medium band (1-2 keV), and 7 were in the hard band (2-7 keV). The count rate, including the PSF correction, equals $0.56^{+0.42}_{-0.29} \times 10^{-3}$ ct s^{-1} (90% confidence). Although this X-ray spectrum (Figure 3) is rather ratty due to the low photon count, the lack of flux below 2 keV implies a high intrinsic absorption ($N_{\rm H} > 10^{22}$ cm^{-2}), and thus a rather intrinsically bright source once corrected. With a Galactic column density of neutral atomic hydrogen $N_{\rm H,Galaxy} = 1.47 \times 10^{20}$ cm^{-2} (HI4PI Collaboration et al. 2016), the unabsorbed luminosity $L_{0.5-8\,\rm keV} \approx 6 \times 10^{38}$ erg s^{-1} despite the small total photon count rate. A summary is provided in Table 1. Details on the data reduction process are provided in Graham et al. (2021) and Soria et al. (2021, in preparation).

2.5 XMM-Newton data

From our analysis of *XMM-Netwon* EPIC-MOS exposures taken in 2010 and 2017 (with 15″ spatial resolution), we find that — after excluding the data affected by particle background flaring (for example solar flares) — the spectrum is best described with two hot-gas components at 0.2 and 0.7 keV (which is fairly standard in star-forming galaxies) plus a power-law component with a photon index $\Gamma = 1.8 \pm 0.8$ (for $N_{\rm H,intrin} = 10^{22}$ cm^{-2}). Moreover, given that the X-ray luminosity of the galaxy's spatially-extended, hot gas clouds are likley to be stable on a time frame of decades, and accounting for detector sensitivity chnages over time, we

[1] The sensitivity of the WFC3/IR detector has been remarkably stable over the past decade (Kozhurina-Platais & Baggett 2020), and offers a 1σ photometric accuracy/repeatability of 1.5 per cent (0.015 mag) (Bajaj 2019).

[2] We use the term 'hot' to denote a star cluster with an active black hole, in this case, X-ray active.

[3] Taken from the Sumi language, spoken by a quarter of a million (formerly all tribal) people in Nagaland, the name relates to the Tuluni, *aka* Anni, festival.

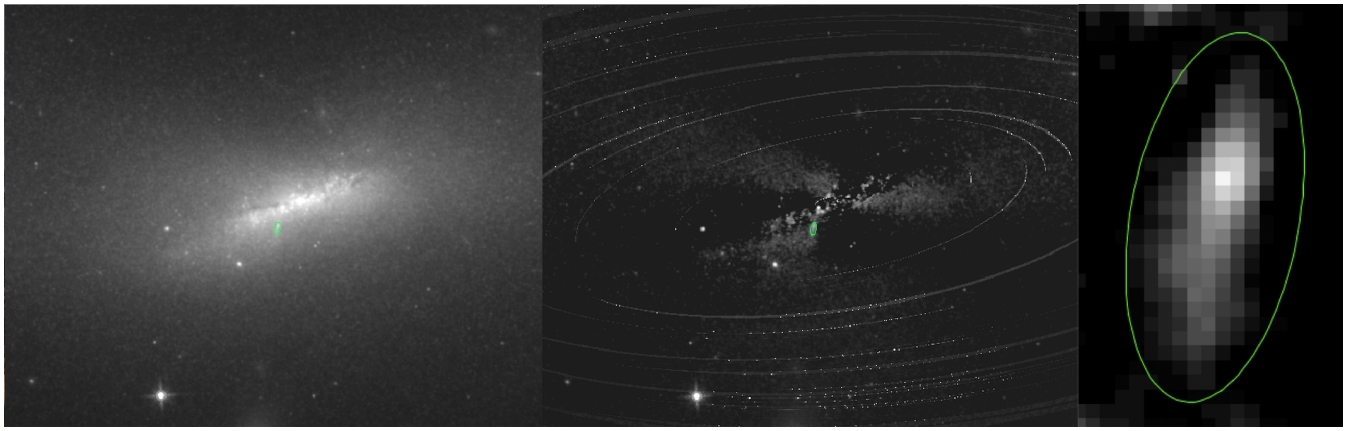

Figure 2. NGC 4424: *HST* WFC3/IR F160W cut-out image (left panel: $78'' \times 61'' = 6.2$ kpc $\times$ 4.8 kpc); Isofit residual image (middle panel); zoom-in of the region containing NGC 4424 X-3 (right panel). The green ellipse is identical in each panel. It is $4''.9$ (390 pc) from the nucleus of NGC 4424, and has a major-axis length of $3''.3$ (260 pc) and a minor-axis length of $1''.5$ (120 pc).

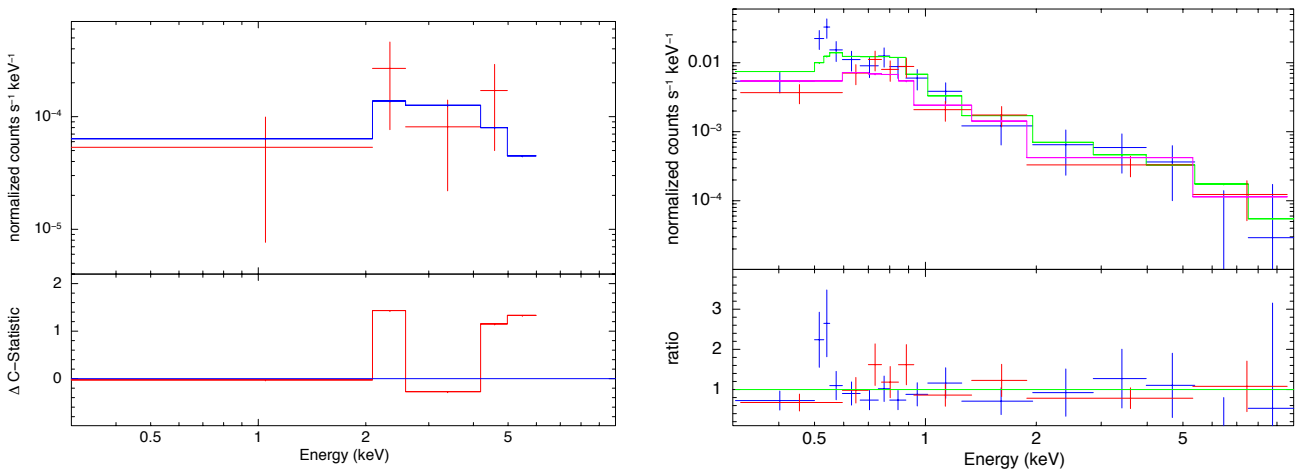

Figure 3. Left panel: CXO/ACIS-S spectrum of NGC 4424 X-3 in Nikhuli, fit with a power-law model. The datapoints have been grouped to a signal-to-noise >1.8 for plotting purposes only. The fit was done on the individual counts, using Cash statistics. Right panel: XMM-Newton/EPIC-MOS spectrum of NGC 4424 X-3 along with two thermal components due to the $15''$ spatial resolution capturing the surrounding hot gas.

Table 1. *CXO* data on NGC 4424 X-3.

$F_{0.5-7\,\mathrm{keV}}$ (mod-indpt) 10^{-14} erg cm^{-2} s^{-1}	$F_{0.5-7\,\mathrm{keV}}$ (mod-dept) 10^{-14} erg cm^{-2} s^{-1}	$L_{0.5-8\,\mathrm{keV}}$ 10^{38} erg s^{-1}	$N_{\mathrm{H,intrin}}$ 10^{22} cm^{-2}	Γ	$L_{0.5-10\,\mathrm{keV}}$ 10^{38} erg s^{-1}	$L_{2-10\,\mathrm{keV}}$ 10^{38} erg s^{-1}
(1)	(2)	(3)	(4)	(5)	(6)	(7)
0.79 (0.39–1.38)	0.89 (0.43–1.64)	6.31 (2.54–13.81)	$2.17^{+3.18}_{-1.64}$	1.7	7.09 (2.85–15.51)	4.57 (1.84–10.01)

Column 1: Observed, i.e. partially-absorbed, model-independent, photon flux in the *CXO*/ACIS-S 0.5–7 keV bands. The associated uncertainty shows the 90% confidence range. Column 2: Absorbed model flux. Column 3: The X-ray luminosity $L_{0.5-8\,\mathrm{keV}}$ represents the unabsorbed 0.5–8 keV luminosity of NGC 4424 X-3, derived using the distance of 16.4 Mpc and corrected for both our Galaxy's obscuring $\mathrm{H_I}$ plus (Column 4) the obscuring line-of-sight $\mathrm{H_I}$ intrinsic to NGC 4424. Column 5: The X-ray SED power-law slope, Γ. Column 6 and 7: Unabsorbed X-ray luminosities $L_{0.5-10\,\mathrm{keV}}$ and $L_{2-10\,\mathrm{keV}}$ based on the extrapolated power-law SED. With $\Gamma = 1.7$, one has that $L_{0.5-8\,\mathrm{keV}} = 1.075\,L_{0.5-7\,\mathrm{keV}}$, and $L_{0.5-10\,\mathrm{keV}} = 1.207\,L_{0.5-7\,\mathrm{keV}}$. Similarly, $L_{2-10\,\mathrm{keV}} = 0.725\,L_{0.5-8\,\mathrm{keV}}$, and $L_{0.5-10\,\mathrm{keV}} = 1.123\,L_{0.5-8\,\mathrm{keV}}$.

Table 2. Black hole mass predictions for NGC 4424.

$\log M_{\rm bh}$ ($M_{*,\rm total}$)	$\log M_{\rm bh}$ (ϕ)	$\log M_{\rm bh}$ (σ)	$\overline{\log M_{\rm bh}}$
4.8±0.8	6.7±0.5[a]	5.0±0.8	4.9±0.6

Black hole masses are in units of solar mass. [a] The pitch angle is considered unreliable for predicting the black hole mass in this post-merger galaxy and is not used here to calculate the error-weighted mean: $\overline{\log M_{\rm bh}}$.

determine that NGC 4424 X-3's 0.3-10 keV X-ray luminosity has changed from 10^{39} erg s^{-1} (2010-Jun-13) to 2.2×10^{39} erg s^{-1} (2017-Dec-5).

3 BLACK HOLE MASS ESTIMATES

NGC 4424 may harbour two massive black holes, possibly with one yet to settle to the centre of this merged system (Merritt & Milosavljević 2005; Barausse 2012; Tremmel et al. 2018), thereby explaining NGC 4424 X-3 in the elongated, off-centre star cluster.

Using our revised distance modulus of 31.080±0.292 (D= 16.4 ± 0.8 Mpc), rather than 30.6±1.0 (Graham et al. 2019, their Appendix), the galaxy NGC 4424 has a stellar mass of $\log M_{*,\rm gal} = 10.0 \pm 0.1$, which implies a central black hole mass of $\log M_{\rm bh} = 4.8 \pm 0.8$ (based upon a symmetrical Bayesian analysis of the $M_{\rm bh}$–$M_{*,\rm gal}$ relation) or 5.1 ± 0.8 (based upon a symmetrical Bisector regression using the modified FITEXY routine), see Davis et al. (2018) and Graham et al. (2019) for details.

The past/ongoing merger event in NGC 4424 has undoubtedly distorted the spiral pattern (Kenney et al. 1996; Cortés et al. 2006), likely accounting for the discrepant prediction, see Table 2, of the black hole mass based on the spiral arm pitch angle (16.9±2.4 degrees: Graham et al. 2019) and the $M_{\rm bh}$–ϕ relation in Davis et al. (2017). The merger may have also elevated the central stellar velocity dispersion, reported to be 57.0 ± 8.6 by Ho et al. (2009), although perhaps this is in line with the increase to the stellar mass as it results in a consistent prediction for the black hole mass of $\log M_{\rm bh} = 5.0 \pm 0.8$ (Davis et al. 2017).

The error-weighted mean of the logarithm of the estimated black hole masses can be calculated using the expression

$$\overline{\log M_{\rm bh}} = \frac{\sum_{i=1}^{N} w_i \log M_{{\rm bh},i}}{\sum_{i=1}^{N} w_i}, \quad (1)$$

with inverse-variance weighting[4] such that $w_i = 1/(\delta \log M_{{\rm bh},i})^2$. The associated 1σ standard error is given by

$$\delta\,\overline{\log M_{\rm bh}} = \sqrt{\frac{1}{\sum_{i=1}^{N} w_i}}. \quad (2)$$

Given that NGC 4424 is predicted to possess an intermediate mass black hole with $\overline{\log M_{\rm bh}} = 4.9 \pm 0.6$ dex (see Table 2), the central black hole of the infalling, nearly-consumed galaxy might be smaller still, and thus also of intermediate-mass (10^2–10^5 M$_\odot$). Graham (2020) has found that the $M_{\rm bh}$–$M_{\rm nsc}$ relation for nuclear star clusters also applies to ultracompact dwarf (UCD) galaxies, thought to be the tidally stripped nuclei of galaxies (Zinnecker et al. 1988; Freeman 1993). From a visual inspection of the (residual) image of NGC 4424, it is hard to establish how much of the infalling nuclear star cluster may have been stripped away, or instead, how much of the infalling galaxy still remains around the infalling nuclear star cluster. Nonetheless, using equation 7 from Graham (2020), and assigning a factor of 2 uncertainty to the stellar-mass — dominated by the uncertainty in the M_*/L_{F160W} ratio — one obtains a (poorly constrained)[5] value of $\log(M_{\rm bh}/{\rm M}_\odot) = 4.8^{+1.8}_{-1.4}$, with an optimal value of $M_{\rm bh} = 6 \times 10^4$ M$_\odot$. This should not be confused with the above estimate for the *central* massive black hole in NGC 4424. For reference, due to the steep (2.62) slope of the $M_{\rm bh}$–$M_{\rm nsc}$ relation, a factor of 2 reduction to the star cluster mass results in a 0.8 dex decrease to the predicted black hole mass.

Using $L_X = L_{0.5-10\,\rm keV} = 7.09 \times 10^{38}$ erg s^{-1} (Table 1), the Eddington ratio ($L_X/L_{\rm Edd}$) of NGC 4424 X-3 can be calculated using $L_{\rm Edd} = 1.3 \times 10^{38} M_{\rm bh}/{\rm M}_\odot$ erg s^{-1}, where $M_{\rm bh} = 6 \times 10^4$ M$_\odot$. This gives an Eddington ratio of 9×10^{-5}, or roughly 10^{-4}. Such a value is consistent with the values seen for IMBH candidates at the centres of early-type galaxies in the Virgo cluster (Graham & Soria 2019, their Figure 8).

4 DISCUSSION

The phenomenon of an infalling galaxy undergoing assimilation has also been offered to explain the offset IMBH in ESO 243-49 (Farrell et al. 2009; Webb et al. 2017). Another interesting example can be seen in the Virgo cluster galaxy NGC 4651 (Martínez-Delgado et al. 2010; Foster et al. 2014; Morales et al. 2018), where the elongation has produced a tidal stream which is immediately evident in the optical images. An earlier stage of such shredding — of an infalling dwarf galaxy — has been reported in a galaxy at a redshift $z = 0.145$ by Forbes et al. (2003), see also UGC 10214 (Vorontsov-Velyaminov 1959; Miskolczi et al. 2011) and NGC 7714 (Soria & Motch 2004). In addition to M32 giving itself to M31 (Graham 2002), various stages of the interaction process can also be seen in the catalog of Vorontsov-Velyaminov (1977), and the process by which a galaxy is threshed and reduced to its dense core has been advanced in a series of papers (Bekki et al. 2001, 2003; Bekki & Freeman 2003). In our own galaxy, Ibata et al. (2019a) reveal the impressive tidal stream associated with ω-Centauri, possibly the pared-core of an accreted dwarf galaxy but currently lacking confirmation of a central black hole (Zocchi et al. 2019; Baumgardt et al. 2019), while Pfeffer et al. (2020) report on other remnant nuclei in the Milky Way. The slightly more massive ultra compact dwarf galaxies (e.g. Ahn et al. 2017), which have also been caught in the act of stripping (e.g. VUCD3: Liu et al. 2015), are known to contain

[4] This gives the 'maximum likelihood estimate' for the mean of the probability distributions under the assumption that they are independent and normally distributed with the same mean.

[5] The heightened uncertainty is due to the steep slope of, and scatter about, the $M_{\rm bh}$–$M_{\rm nsc}$ relation.

massive black holes (Graham 2020, and references therein), and are probably fertile grounds for finding IMBHs.

An incoming black hole of equal mass to that already (presumed to be) at the centre of NGC 4424 might imply a major merger between two spiral galaxies, which would be unlikely to result in a post merger galaxy looking like a spiral galaxy. However, a lower mass early-type galaxy may well have been consumed, as they have smaller stellar masses for the same black hole mass. Such a merger also meshes with the notion of an unequal mass merger reported by Boselli et al. (2018). From Figure 11 in Sahu et al. (2019), one can see that an early-type galaxy with a black mass of $10^{4.8}$ $M_\odot$ will have a stellar mass of $\sim 0.6 \times 10^9$ $M_\odot$, which is just 6% of the stellar mass of NGC 4424. If such an early-type galaxy was what fell into NGC 4424, then (ignoring gaseous processes), the coalesence of the two black holes will double the central black hole mass, with the stellar mass increasing by only ∼6%. We speculate that such minor mergers may help maintain the steepness of the non-linear, near-cubic $M_{\rm bh}$–$M_{*,\rm gal}$ relation (Savorgnan et al. 2016; Davis et al. 2018).

Given the similarity of the predicted (central and infalling) black hole masses, and the absence of an X-ray point source at the centre of NGC 4424 (despite the presence of gas: Boselli et al. 2018), it might (but need not) be that NGC 4424 currently has no central black hole, and the delivery of NGC 4424 X-3 via a low-mass early-type galaxy is how bulgeless late-type galaxies are seeded with black holes. At 6×10^{10} $M_\odot$, the Milky Way has a stellar mass some six times bigger than NGC 4424 (Licquia & Newman 2015). If six minor merger events have brought in six 10^5 $M_{\rm odot}$ black holes, one might (naively) expect a central black hole with a minimum mass of 0.6×10^6 $M_\odot$, with gas fuelling and stellar capture increasing this to the observed value of 4×10^6 M_{odot} (Boehle et al. 2016). It may also explain the tentative suggestions of IMBHs in the Milky Way (Takekawa et al. 2020, and references therein). Although, while Askar et al. (2020) have modelled how the delivery of an IMBH into a pre-existing star cluster can result in its growth into an SMBH, it may also be ejected in the event of a merging binary IMBH with mass ratio $\gtrsim$0.15. The merging process is also known to heat and erode the surrounding star cluster (Bekki & Graham 2010), and thus the simple addition of masses would be simplistic.

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DATA AVAILABILITY

The data underlying this article are available in the Chandra Data Archive (CDA: `https://cxc.harvard.edu/cda/`), the Next Generation Virgo Cluster Survey website (`https://www.cfht.hawaii.edu/Science/NGVS/`), and the Hubble Space Telescope section of the Mikulski Archive for Space Telescopes (MAST: `https://archive.stsci.edu/hst/`).

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